MIL-G-15342E(SH)
5 January 1987
SUPERSEDING
MIL-G-15342D(SH)
5 June 1976
(See 6.5)

MILITARY SPECIFICATION

GASKETS, METALLIC, SPIRAL WOUND FOR BOILERS

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers metallic spiral wound gaskets for handholes and manholes of shipboard and stationary boilers. Steam generators for nuclear plants are excluded.

1.2 Classification. Gaskets shall be of the following classes and series, as specified (see 6.2):

Class 1 - Boiler handhole:
- (a) Series 600-850 degrees Fahrenheit (°F)
- (b) Series 1500-950°F

Class 2 - Boiler manhole:
- (a) Series 600-500°F
- (b) Series 1500-600°F

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5330
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SPECIFICATIONS

FEDERAL

PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-665	- Boxes: Paperboard, Metal Edged and Components.
PPP-B-676	- Boxes, Setup.

MILITARY

MIL-L-10547	- Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

FEDERAL

FED-STD-151	- Metals; Test Methods.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-2073-1A	- DoD Material Procedures for Development and Application of Packaging Requirements.

2.1.2 Government drawing. The following Government drawing forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWING

NAVAL SEA SYSTEMS COMMAND (NAVSEA)
803-5959278 - Gasket, Main Boiler Handhole and Manhole.

(Copies of specifications, standards and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 167	- Standard Specification for Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip. (DoD adopted)
A 240	- Standard Specification for Heat-Resisting Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels. (DoD adopted)
D 512	- Standard Test Methods for Chloride Ion in Water.
D 3951	- Standard Practice for Commercial Packaging.

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT
National Motor Freight Classification

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., ATA TRAFFIC Dept., 2200 Mill Road, Alexandria, VA 22314.)

TECHNICAL ASSOCIATION OF THE PULP AND PAPER INDUSTRY (TAPPI)
T256-Water-Soluble Chlorides in Pulp and Paper.

(Application for copies should be addressed to the Technical Association of the Pulp and Paper Industry, 1 Dunwoody Park, Atlanta, GA 30338.)

UNIFORM CLASSIFICATION COMMITTEE AGENT
Uniform Freight Classification Ratings, Rules and Regulations

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Metallic spiral wound gaskets furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.3).

3.2 Construction. Gaskets shall be composed of a single thickness of metal wound in a flat spiral shape, starting on the inside and working outward ply by ply to the desired size. Between the plies of metal shall be a cushion of filler material wound to conform with the metal strip. For class 1 and 2 gaskets, spot welds, shall be as shown on Drawing 803-5959278.

3.2.1 Dimensions and tolerances. Unless otherwise specified (see 6.2), gaskets shall conform to the dimensions, tolerances and other requirements as shown on Drawing 803-5959278. Class 2 gaskets shall be elliptical in shape.

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3.2.2 Plies. A ply shall consist of one 360-degree turn of the metal or filler strip. Gaskets shall be wound on a mandrel of the desired shape. After one complete revolution, the initial spot welds shall be made. Approximately midway during the next revolution, the filler strip shall be introduced. Winding shall continue to make the product contain the number of plies of filler required. A minimum of two plies of metal shall then be made and then the terminal spot welds shall be made.

3.3 Materials.

3.3.1 Metal strip. The metal strip used in the gasket shall be one of the following materials:

(a) Corrosion-resisting steel (UNS S34700), cold rolled and annealed in accordance with ASTM A 167, type 347, or ASTM A 240, type 347.
(b) Corrosion-resisting steel (UNS S31603), in accordance with ASTM A 167, type 316L, or ASTM A 240, type 316L.
(c) Corrosion-resisting steel, (UNS S30908) in accordance with ASTM A 167, type 309S, except that type 309S shall be stabilized with columbium (and tantalum not exceeding 0.10 percent) and their combination to be not less than 10 times the carbon content.

3.3.1.1 Thickness. The thickness of the metal strip shall be between 0.0063 and 0.0088 inch.

3.3.2 Filler material. The filler material shall be in paper strip form and shall be in accordance with Drawing 803-5959278. The material shall be selected for low chloride content (see 3.3.2.1). The filler material shall contain less than 500 parts per million (p/m) of zinc and shall not contain the following:

(a) Grease, lead or lead components, mercury or mercury compounds.
(b) Asbestos fibers and components containing asbestos fibers.

3.3.2.1 Chloride content of filler. The chloride content of the filler material shall meet the following requirements, when tested as specified in 4.6.2.1 and 4.6.2.2:

(a) Water leachable - Not greater than 220 p/m.
(b) Total chlorides - Not greater than 550 p/m.

3.3.3 Recovered materials. Unless otherwise specified herein, material and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

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3.4 Service conditions.

3.4.1 Service conditions and temperature ratings. Gaskets shall be rated for service at the temperatures specified in 1.2.

3.4.2 Service condition of bearing surface. Gaskets shall be rated for service in connection with flat smooth surfaces having finishes of spiral or concentric lay of 60 to 80 serrations per inch and a roughness height rating (RHR) of 125.

3.5 Cyclic and hydrostatic test. There shall be no visible steam or water leakage from the flanged joint after the first five cycles of the service test or during the hydrostatic test, when tested as specified in 4.6.3. Upon completion of the tests, the gasket shall remain intact (metal strip and spot weld unbroken).

3.6 Marking. Unless otherwise specified (see 6.2), gaskets shall be marked with a sturdy paper tag (of tag stock) securely attached thereto, which shall contain thereon the following information: The manufacturer's name; gasket size, including shape, thickness, inside and outside dimensions; type of metal and filler material; pressure and temperature rating, national stock number, and customer's order number. For class 1, at least one tag shall be applied to each 25 gaskets or less. For class 2 gaskets, each gasket shall be tagged.

3.7 Workmanship. The finished gaskets shall have no breaks, creases or sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Qualification inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

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4.3 Qualification tests. Qualification tests shall be conducted at a laboratory satisfactory to NAVSEA. Qualification tests shall consist of the examinations and tests specified in 4.5 and 4.6. Contractors shall furnish a minimum of six representative samples for each class and series of gasket for which qualification is desired.

4.4 Quality conformance inspection. Each of the sample gaskets selected in accordance with 4.4.1.2 shall be analyzed in accordance with method 111 of FED-STD-151 to determine compliance with 4.5. If any sample fails to conform to 4.5, this shall be cause for rejection of the entire lot.

4.4.1 Sampling.

4.4.1.1 Lot. Gaskets of the same size and series manufactured from the same batch of material and presented at one time shall be considered a lot for purposes of examination and tests.

4.4.1.2 Sampling for examination. A random sample of gaskets shall be selected from each lot of material offered for inspection of visual and dimensional characteristics in accordance with the sampling procedures of MIL-STD-105, inspection level S-4. Acceptable quality level (AQL) shall be 1.5 percent defective.

4.4.1.3 Sampling for chemical analysis.

4.4.1.3.1 Metal strip. Five feet of uncrimped metal strip representing the lot shall be forwarded to the testing facility for purposes of chemical analysis of metal.

4.4.1.3.2 Filler. Fifty grams of the filler paper representing the lot shall be forwarded to the testing facility for purposes of chemical analysis.

4.5 Visual and dimensional examination. Each of the sample gaskets selected in accordance with 4.4.1.2 shall be visually and dimensionally examined to verify compliance with this specification. Any gasket in the sample containing one or more defects shall not be offered for delivery, and if the number defective exceeds the acceptance number, this shall be cause for rejection of the lot represented by the sample. Examination shall be conducted as specified in table I.

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TABLE I. Classification of defects.

Categories	Defects
Critical	None defined.
Major:	
101	Class of gasket not as specified.
102	Rating of gasket not as specified (pressure series).
103	Thickness of finished gasket not within specified tolerances.
104	Dimensioning of gasket not as specified in contract or on standard drawing.
105	Unnecessary breaks, creases or sharp edges on finished gaskets.
106	Identification markings missing on tags, illegible, incorrect, or not as specified.
107	Packaging or packing not as specified.
108	Number of metal plies less than specified minimum.
109	Number of filler plies less than minimum specified, or plies less than a 360 degrees each where specified.
110	Spot welds missing, not minimum number, incorrectly spaced or defective.
111	Thickness of metal strip not within specified minimum-maximum, or not continuous.
Minor	None defined.

4.6 Tests.

4.6.1 Chemical analysis of metal strip. The metal strip selected in compliance with 4.4.1.3.1 shall be analyzed in accordance with method 111 of FED-STD-151 to determine compliance with 3.3.1 and 3.3.1.1. If any sample fails to conform, this shall be cause for rejection of the entire lot.

4.6.2 Chemical analysis of filler material. The filler paper selected in compliance with 4.4.1.3.2 shall be analyzed to determine compliance with 3.3.2 and 3.3.2.1. Test for leachable chloride shall be as specified in 4.6.2.1. Test for total chloride shall be as specified in 4.6.2.2. If any sample fails to conform, this shall be cause for rejection of the entire lot.

4.6.2.1 Water leachable chlorides. For determination of water leachable chloride, the following test method shall be used:

(a) Two test specimens, each of approximately but not less than 15 grams, shall be taken. Each test specimen shall be taken from a different sample. The test specimen of filler material shall be representative of the sample cross section.
(b) The test specimens shall be cut into pieces no larger than 1/4 by 1/2 inch.

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(c) Each test specimen shall be weighed, transferred to Erlenmeyer flask, equipped with a reflux condenser, covered with 200 to 300 milliliters (mL) of distilled or demineralized water and simmered in the range of 200 to 212°F for a minimum of 6 hours. The leach water shall then be separated by filtration and the filter rinsed. A blank determination shall be run using similarly cleaned equipment and distilled or demineralized water from the same source.

(d) The extracted chloride ion shall be determined by the mercurimetric titration method as specified in ASTM D 512, method A, and shall be corrected by the results of the blank determination.

(e) The results of each test shall be calculated as net p/m of extractable chloride ion weight of the test specimen and shall meet the requirements as specified in 3.3.2.1.

4.6.2.2 Total chlorides. A specimen from one sample shall be tested for chloride in accordance with TAPPI T256. If the test results exceed the requirement in accordance with TAPPI T256, specimens from two other samples may be tested by the same procedure, and the average of the three tests shall be as specified in 3.3.2.1.

4.6.3 Cyclic pressure and temperature test. The test shall be conducted utilizing a simulated boiler closure and a test fixture capable of producing and maintaining superheated steam conditions of an average temperature of 950°F at a pressure of 1275 pounds per square inch (lb/in^2). A gasket in accordance with class 1, series 1500 as specified herein shall be installed in the simulated boiler closure. Thermocouples shall be installed to indicate the internal pipe wall temperature. Prior to testing, a hydrostatic test at 2070 lb/in^2 (150 percent) shall be applied to the fixture. The test procedure consists of five pressure cycles during an 8-hour period. This shall be followed by a 16 to 72 hour period where the test conditions are reduced to 0 lb/in^2 and 200°F for the header metal temperature. A pressure cycle is defined as 0 lb/in^2 to 1275 lb/in^2 to 0 lb/in^2 at a steam temperature of 950°F. The entire test shall consist of 150 pressure cycles with thermal cycles after each five pressure cycles as specified herein. At the completion of the test, a hydrostatic test at 2070 lb/in^2 (150 percent) shall be applied to the fixture. The service test and hydrostatic test results shall meet the requirements in 3.5.

4.7 Comparison inspection. Comparison inspection shall be conducted to verify continuing quality of the approved product and compliance with the specification requirements. These tests will be on samples representing those series and sizes which are tested for qualification and approved, and will be conducted on six samples of each series and size, supplied by the contractor at no cost to the Government. Comparison inspection is not required more often than every 2 years unless the Government has reason to consider it necessary. Comparison examination and tests shall be those specified in 4.5, 4.6.1 and 4.6.2 plus any others specifically requested by the Government.

4.8 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

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5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 Preservation. Preservation shall be level A, C, or commercial as specified (see 6.2).

5.1.1 Level A. Gaskets, quantity as specified (see 6.2), shall be unit protected in boxes in accordance with PPP-B-566, PPP-B-636, PPP-B-665, PPP-B-676, at the option of the contractor. Box closure shall be as specified in the applicable box specification or appendix thereto.

5.1.1.1 Bundling. When specified (see 6.2), gaskets shall be bundled in quantities, tied, and wrapped in a noncorrosive, dust protective opaque or transparent paper or film as follows:

Class 1 - 25 gaskets per bundle
Class 2 - 12 gaskets per bundle

The wrapper shall be secured with pressure sensitive tape or heat sealed.

5.1.2 Level C. Gaskets shall be preserved as specified for level A, except that unit containers may be of the non-weather resistant, domestic type or class.

5.1.3 Commercial. Gaskets, quantity as specified (see 6.2), shall be unit packed in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, C, or commercial as specified (see 6.2).

5.2.1 Levels A, B, and C. Gaskets preserved as specified in 5.1 shall be packed in containers conforming to the exterior shipping container tables listed in MIL-STD-2073-1A for the level of packing specified.

5.2.2 Commercial. Gaskets preserved as specified in 5.1 shall be packed in accordance with ASTM D 3951.

5.2.3 Filling of voids. Voids or cavities in the shipping containers shall be filled with cushioning materials to prevent shifting of contents during handling, shipment and storage. The use of loose excelsior, shredded newspaper and similar materials which have not been refined for noncorrosiveness shall not be used unless these materials were enclosed in a sealed waterproof barrier material.

5.3 Marking. In addition to any special marking required by the contract or order (see 6.2), interior packages and exterior shipping containers for levels A, B, and C shall be marked in accordance with MIL-STD-129, commercial packs in accordance with ASTM D 3951.

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6. NOTES

6.1 Intended use. The metallic spiral wound gaskets covered by this specification are intended for use in handholes and manholes of shipboard and stationary boilers with temperatures and pressures specified (see 1.2).

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Class and series of gasket (see 1.2).
(c) Line item number for gasket from standard drawing, or if not listed, a full description such as where used in boiler, inside and outside dimensions, thickness, bearing surface and all other pertinent information (see 3.2.1 and 3.6).
(d) Level of packaging and packing (see 5.1, 5.2 and 5.2.2).
(e) Quantity per unit package (see 5.1.1).
(f) When bundling is required (see 5.1.1.1).
(g) When use of expanded, loose-fill polystyrene is approved (see 5.2.4).
(h) Special marking, if required (see 5.3).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in Qualified Products List QPL-15342 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 and information pertaining to qualification of products may be obtained from that activity. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3.1).

6.3.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

6.4 Subject term (key word) listing.

Boiler handhole
Boiler manhole
Gaskets
Metal strip

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 5330-N089)

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